

Navigating Borders and Identities: Aligning Governance and Immersive Design for Public Sector Engagement in the Metaverse

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ABSTRACT

As public entities enter the metaverse, their efforts to meet public service obligations and uphold ethical values lead to governance challenges that directly affect immersive design. Metaverse characteristics of anonymity, interoperability, borderlessness and immersivity create tensions that need to be addressed. Drawing on interviews and document analysis across three metaverse initiatives, this study identifies key governance concerns around jurisdictional boundaries, regulatory alignment, and identity management. These insights inform an emerging framework for ethically aligned and compliant public sector engagement in the metaverse.

Index Terms: Metaverse, ethical governance, immersive, public sector.

1 INTRODUCTION

The public sector (PS) plays a crucial role in how society functions and how citizens navigate daily life. As other industries adopt emerging technologies, the PS also adapts to the expectations of a digital society [1]. However, its mandate to uphold values and serve the needs of the public introduces added considerations for engaging with citizens in the metaverse [2]. The metaverse is an ecosystem of immersive spaces where users interact as avatars across a network of interoperable digital technologies [3, 4]. Its features of immersivity, anonymity, interoperability, and borderlessness challenges the PS's ability to fulfil societal mandates [5].

Senadheera et al. [6], outlined five indications of PS obligations, including greater public transparency, increased accountability, prevalence of red tape, intensive regulatory environment, and extensive political oversight. These, along with context-specific regulation shape the sector's entry and service delivery in metaverse ecosystems. These requirements influence micro-level user experience fundamentals, such as presence, embodiment, privacy, and interaction fidelity. Ethical misalignment can discourage adoption and undermine beneficial innovation. In Europe, requirements are established through regulations such as GDPR and the Artificial Intelligence (AI) Act that promote privacy, safety, transparency, and accountability, which are key obligations of the PS. Understanding PS governance

is essential for Virtual Reality (VR) researchers working on immersive technologies. For instance, identity management requirements need to be considered without compromising immersion or privacy, regulatory jurisdiction may affect how presence is designed, and accountability requirements may dictate specific interaction-related data flows.

Pioneering initiatives like Norway's Business Register Office and Tax Administration presence in Decentraland and South Korea's Metaverse Seoul, a virtual world where users can engage with city services and socialise, highlight the complexities of PS entering the metaverse [7]. Furthermore, the Citiverse Initiative [8] aims to address these issues through strategic governance alignment across global virtual cities. Collectively, these efforts offer empirical insights into how ethical alignment is achieved and how compliance is actualised in real-world contexts, helping to inform the adaptation of governance mechanisms. Foregrounding ethical alignment and compliance for effective metaverse governance and immersive system design, our study examines the following research question:

What governance mechanisms support ethically compliant public sector engagement in the metaverse?

2 METHOD

The study employs an interpretative qualitative case study approach [9], to investigate how metaverse governance is shaped to meet PS's ethical and compliance needs. Data collection consists of interviews with stakeholders involved in the Norwegian, Seoul, and Citiverse initiatives, providing real-world insights into PS engagement in the metaverse. Interviews were complemented by the analysis of policy, governance, and strategy documents for each initiative. Currently, seven interviews were conducted with metaverse industry experts and PS innovation and marketing leads, alongside the analysis of over ten documents, with additional data forthcoming. Interviews focused on background knowledge, metaverse design, and governance considerations, and were transcribed and reviewed by the research team. Data were analysed following a grounded theory approach [10], to identify codes (e.g. 'user identity', 'virtual border clarity') and themes.

3 PRELIMINARY FINDINGS

Extending earlier insights on PS obligations in the metaverse [1, 6], this study explores how governance structures operate in practice as public entities engage with metaverse ecosystems. Initial analysis identifies three interrelated concerns: **defining jurisdictional boundaries, aligning with regulation, and ensuring robust identity management**. These form the early foundation of a conceptual governance framework that aligns institutional responsibilities with governance design and the experiential dynamics of immersive systems.

Across all three cases, stakeholders emphasised that the metaverse's borderlessness creates uncertainty around accountability and the enforceability of public obligations. PS

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actors require clearly delineated jurisdictions to determine where authority begins and ends and which regulatory regimes apply. Empirically, actors described difficulties in determining which “space” they legally occupy when delivering services across interconnected virtual worlds. This raises concerns about liability, legitimacy of public-service interactions, and citizens' expectations for due process. Immersive environments may need explicit cues (such as visual demarcations, signage, or governance overlays) to signal entry into regulated public-service spaces. Transitions between zones may require design patterns that maintain presence while communicating governance boundaries. These insights show that presence is not only perceptual but institutionally situated: users must know which regulations apply to feel safe, oriented, and informed.

A second concern relates to transferability of existing regulation, such as GDPR or national administrative law, into immersive environments. This is important because PS actors must demonstrate that immersive interactions meet regulatory standards. Immersive systems may require embedded transparency mechanisms, such as contextual consent prompts, or explainability cues that do not disrupt presence. Regulatory alignment ultimately shapes interaction fidelity, as some interactions may require redesign to remain compliant while still feeling natural.

A third concern involves the need for reliable identity management when engaging with public services. PS obligations require secure, verifiable identity for safeguarding against fraud and ensuring fair access. Identity is both a technical and ethical challenge: authentication must satisfy regulatory requirements while remaining unobtrusive enough to avoid undermining immersion and preserving privacy. This also requires some regulatory alignment. VR systems may need to support layered identity models, enabling users to retain pseudonymous social presence while selectively revealing verified identity for specific public-service interactions.

Figure 1 presents how metaverse features create tensions around accountability (AC) and transparency (TP) when PS actors aim for defined jurisdictional boundaries, regulatory alignment, and robust identity management.

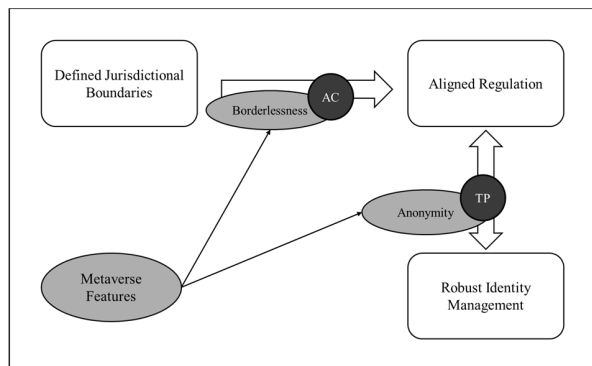


Figure 1: Metaverse features and tensions with governance components

These findings highlight that governance is not an external constraint, it is a design parameter for immersive systems. They highlight opportunities for VR researchers to explore new design frameworks that integrate governance considerations into immersive user experiences.

4 CONCLUSION

Public sector entry into the metaverse raises governance challenges that lie at the intersection of immersive design, ethics, and

regulation. By examining real-world initiatives, this study identifies governance components (related to jurisdictional boundaries, regulatory alignment, and identity management) that directly influence the design and user experience of immersive environments. This study advances the ethical and sociocultural perspectives on governance by examining how the metaverse intersects with PS obligations; an area still largely unexplored in existing literature. While prior research offers valuable theoretical contributions [5, 6], this work introduces empirical evidence drawn from real-world cases to inform the development of ethically aligned and compliant governance structures and introduces design-relevant constraints rather than purely administrative concerns. It foregrounds the importance of ethical and regulatory alignment for technology adoption and expansion to new use cases. Initial findings reveal key governance components for PS actors, as well as the tensions that arise due to metaverse features. These tensions underscore the need for VR researchers to integrate governance considerations into the development of immersive environments.

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